Database overview

Dubai Real Estate CRM · Document 03 · Version 1.2 · 11 October 2026 · For stakeholders · Full technical version: CRM_Database_Design.pdf

77	230	51/51	UAE
tables (lists of records)	links between them	safety rules tested and passed	stored in AWS UAE, backed up daily

What the database is, in simple words

The database is the CRM's **filing cabinet**. Every lead, call, message, unit, deal, invoice and check-in is kept here as a record in a table, much like a row in Excel. Tables are linked: a lead belongs to an agent, a deal points to one unit, an invoice belongs to a deal. The most important business rules are built into the database itself, so they hold even if the app has a bug.

Each company's data is separate

Every record carries its company. One brokerage can never see, change or link to another brokerage's data.

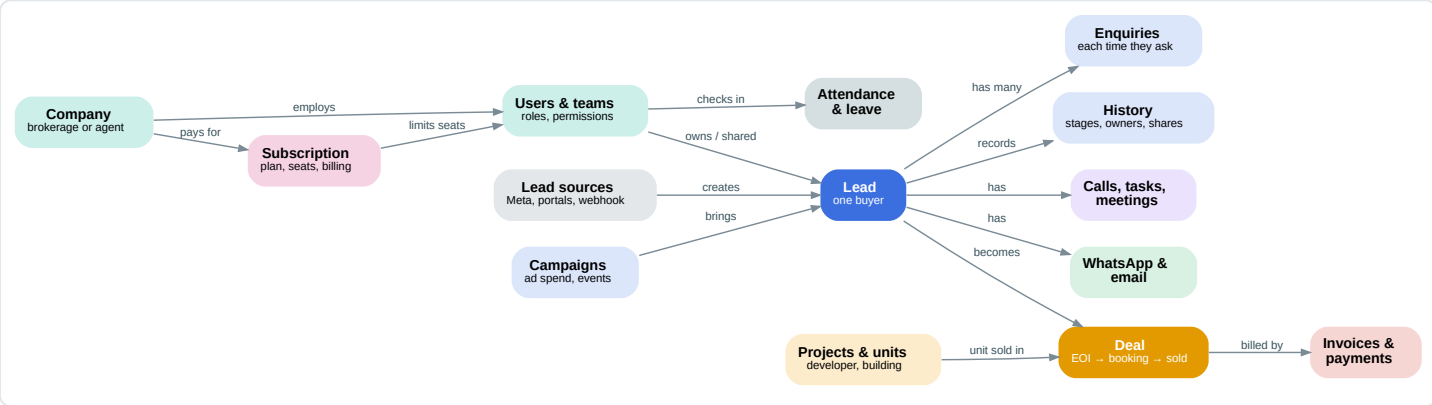
Nothing is silently lost

History (stage changes, owner moves, shares, enquiries, audit log) can only be added to, never edited or deleted. Leads are never auto-deleted.

Rules the app cannot break

No double booking, no double import, no unit sold twice, a reason required for Lost, and invoice amounts locked after approval.

The big picture



Simplified: each box is a group of tables. The lead is at the centre; a deal links one lead to one unit and leads to invoices. The subscription decides how many users a company can have.

The 10 areas at a glance

Organization, roles and settings	11	Each customer company (tenant) has its own users, teams and roles. Roles decide what each person can do; Admin can create new roles such as HR.
Leads	11	Every buyer is one lead. Each time they enquire, an enquiry row is added (repeat enquiries go to the same lead and agent). Every stage change, owner change and share is recorded and can never be edited.
Lead assignment and marketing	5	Rules decide which agent gets a new lead. Campaigns, daily ad spend and events tell us where leads came from and what they cost.
Tasks, meetings and notifications	6	Calls, follow-ups, meetings and viewings. Each agent has a booking page so clients can book themselves. The database itself blocks double booking.
WhatsApp and email	6	All WhatsApp and email messages with their delivery status, approved templates, bulk campaigns and each lead's consent history.
Projects, units and deals	6	Developers, projects, buildings and units for sale, with every price and status change kept. A deal links a buyer to one unit; a unit can only be booked once.
Finance	7	Invoices and purchase orders with lines, VAT, payments and full approval history. After approval, amounts are locked by the database.
Attendance and leave	6	One attendance row per person per day with check-in photo. Corrections keep the original values. Approved leave cannot overlap and blocks new leads.
Subscriptions, billing and owner console	11	How we sell the CRM. Each customer company has one subscription: plan, number of paid users (seats), monthly or yearly, card or bank transfer. We send VAT invoices; Stripe confirms card payments and the platform team confirms bank transfers. If payment fails, the company gets 14 days, then becomes read-only; data is never deleted while unpaid. The owner console is only for the platform team.
Integrations, files, reports and audit	8	Connected accounts (Meta, portals, WhatsApp, Google, Microsoft, webhook), every incoming webhook, sync runs, Meta reconciliation, files, exports and the audit log.

Safety rules proven on a real database (51 of 51 passed)

- ✓ Tenant A user sees only tenant A leads
- ✓ No tenant set: sees nothing
- ✓ Tenant A cannot insert a row for tenant B
- ✓ Tenant A cannot read tenant B's lead by ID
- ✓ Lead cannot point to another tenant's user
- ✓ Same submission cannot be imported twice
- ✓ Stage category must match the stage
- ✓ Lost requires a reason
- ✓ Lost with a reason is accepted
- ✓ Open assigned lead must have a next follow-up
- ✓ Response deadline only when the lead has an owner
- ✓ 10-minute rule move is logged (no_response)
- ✓ Admin fallback is logged
- ✓ Manual reassignment needs a reason

- ✓ Lead can be shared as 'edit'
- ✓ Share access must be view or edit
- ✓ Agent cannot be double-booked
- ✓ Booking-page meeting must reference a booking page
- ✓ Zoom is no longer an allowed channel
- ✓ Deal unit must belong to the deal's project
- ✓ A unit cannot be booked by two deals
- ✓ Invoice total and balance are calculated
- ✓ Draft invoice cannot be 'paid' without approval
- ✓ Approved invoice amounts are frozen
- ✓ Lines of an approved invoice cannot change
- ✓ Purchase order cannot be 'paid'
- ✓ History is append-only (audit log)
- ✓ Enquiries cannot be edited
- ✓ Overlapping approved leave is blocked
- ✓ Nobody approves their own leave
- ✓ A file attaches to exactly one record
- ✓ Only Admin has export permissions (default)
- ✓ Admin can switch off lead assignment for a user
- ✓ Active users cannot exceed paid seats
- ✓ Invited (not active) users don't use a seat
- ✓ Seats cannot drop below active users
- ✓ Admin can add seats
- ✓ Tenant cannot change its own billing status
- ✓ Tenant cannot mark itself verified
- ✓ Tenant cannot create its own invoices or payments
- ✓ Tenant sees only its own subscription invoices
- ✓ Trial needs a card on file
- ✓ Payment failure needs a grace end date
- ✓ Cancelled subscription has a data deletion date
- ✓ Bank transfer counts only when confirmed by the platform team
- ✓ Invoice VAT total is calculated (AED 396 + 5%)
- ✓ Issued subscription invoice cannot be edited
- ✓ Brokerage needs trade licence and RERA ORN
- ✓ Individual agent needs RERA broker card (BRN)
- ✓ Subscription history is append-only
- ✓ App cannot see platform admins

Each area: what it stores, with an example

1 Organization, roles and settings

11 tables

Each customer company (tenant) has its own users, teams and roles. Roles decide what each person can do; Admin can create new roles such as HR.

Example: “Skyline Realty” is one company. Sara is a Sales agent in Team Marina; her role lets her see her team's leads only. HR staff get an HR role with no access to leads.

`tenants`	One customer account: a brokerage (B2B) or an individual agent (B2C).
`roles`	Roles per tenant (Director, GM, Sales Manager, Team Leader, Agent, Admin, Finance) and their data scope.
`role_permissions`	Which actions each role may perform.
`permissions`	Catalogue of actions that can be granted to roles.
`users`	People who log in. Each user belongs to one tenant and has one role.
`divisions`	Business divisions, each run by one General Manager.
`teams`	Sales teams. A Sales Manager can manage several teams; each team has one Team Leader.
`tenant_settings`	Tenant configuration values (response time, auto-close policy, scoring weights, work hours, etc.).
`tenant_currencies`	Currencies enabled for a tenant.
`exchange_rates`	Exchange rates to the tenant base currency, used for report totals.
`currencies`	ISO 4217 currencies the system can use.

2 Leads

11 tables

Every buyer is one lead. Each time they enquire, an enquiry row is added (repeat enquiries go to the same lead and agent). Every stage change, owner change and share is recorded and can never be edited.

Example: Ahmed enquires on Bayut about a 2-bed in Dubai Hills, so a lead is created. A week later he enquires again on Property Finder. The new enquiry is added to the same lead, and Sara gets a “Re-enquired” alert.

`leads`	One person interested in property. stage_category is kept in step with the stage by a foreign key.
`lead_inquiries`	Every enquiry received for a lead, with original source IDs and form answers. Never changed after saving.
`lead_projects`	Projects a lead is interested in. One can be marked primary (shown in the list).
`lead_stage_history`	Every stage change (used for the funnel and audit). Never changed after saving.
`lead_assignments`	Every owner change: previous owner, new owner, who, when, why. Never changed after saving.
`lead_shares`	Secondary users who can see (view) or work on (edit) a lead. The owner stays the owner.
`activities`	The lead timeline: one row per event (note, call, message, meeting, stage change, invoice...). Typed links point to the source row.
`saved_filters`	Saved list filters and column choices, private or shared with a team.
`lead_stages`	Configurable lead stages. category drives rules: open, won, lost, invalid.
`reasons`	Configurable reasons for Lost, Invalid and reassignment.
`lead_sources`	Where leads come from (Meta, portals, website, import, events...). quality_score feeds lead scoring.

3 Lead assignment and marketing

5 tables

Rules decide which agent gets a new lead. Campaigns, daily ad spend and events tell us where leads came from and what they cost.

Example: Leads from the “Emaar Beachfront” campaign go to Team Marina in turn. If Sara does not respond in 10 minutes the lead moves to Omar, then Lina, then the Admin, and every move is recorded.

`assignment_rules`	Rules checked top to bottom to choose the owner of a new lead.
`assignment_rule_members`	Specific agents a rule rotates between (used when no target team is set).
`campaigns`	Ad campaigns from Meta, Google, etc., or manual campaigns.
`ad_spend_daily`	Daily ad spend per campaign, imported from ad accounts or entered manually.
`events`	Events and roadshows that leads can be tagged to.

4 Tasks, meetings and notifications

6 tables

Calls, follow-ups, meetings and viewings. Each agent has a booking page so clients can book themselves. The database itself blocks double booking.

Example: Ahmed books a viewing on Sara's booking page for Thursday 4 pm. Nobody else can book Sara for that time.

`tasks`	Calls, follow-ups, meetings, viewings and personal tasks with reminders and outcomes.
`meetings`	Online meetings (Google Meet / Teams) and property viewings. The database blocks double booking of an agent.
`booking_pages`	Each agent's public booking page (replaces Calendly). Clients pick a free slot from the agent's calendar.
`calendar_event_links`	Links a meeting or task to the event copy in a user's Google / Microsoft calendar.
`notifications`	In-app notifications (reminders, escalations, assignments). Also sent as push.
`push_devices`	Phones and browsers registered for push notifications.

5 WhatsApp and email

6 tables

All WhatsApp and email messages with their delivery status, approved templates, bulk campaigns and each lead's consent history.

Example: A WhatsApp template “New launch in JVC” is sent to 300 leads who agreed to be contacted. Each message shows sent, delivered or read.

`conversations`	One WhatsApp or email thread per lead, channel and sending account.
`messages`	Every WhatsApp and email message, in and out, with delivery status.
`message_templates`	Approved WhatsApp templates and email templates.
`message_campaigns`	Bulk WhatsApp / email campaigns with test, approval and schedule.
`message_campaign_recipients`	Each lead in a bulk campaign and what happened to its message.
`consents`	Consent and opt-out history per lead and channel. Latest row wins. Never changed after saving.

6 Projects, units and deals

6 tables

Developers, projects, buildings and units for sale, with every price and status change kept. A deal links a buyer to one unit; a unit can only be booked once.

Example: Unit 1204 in Tower B is booked under Ahmed's deal. A second deal for unit 1204 is refused by the database.

`developers`	Property developers (e.g. Emaar, Sobha).
`projects`	Real estate projects sold by the brokerage.
`buildings`	Buildings or phases within a project.
`units`	Individual units for sale. Status: available, on hold, booked, sold.
`unit_changes`	History of every unit price and status change.
`deals`	A lead's purchase of one unit: offer, EOI, booking and sale.

7 Finance

7 tables

Invoices and purchase orders with lines, VAT, payments and full approval history. After approval, amounts are locked by the database.

Example: An invoice to the developer for AED 52,500 commission plus 5% VAT. After the manager approves it, the amount can no longer be changed.

`finance_documents`	Sales invoices, purchase invoices and purchase orders.
`finance_document_lines`	Line items of a financial document. Can only change while the document is Draft.
`payments`	Payments and refunds recorded against a document.
`finance_document_events`	Approval and status history of each document. Never changed after saving.
`counterparties`	Who a financial document is with: client, developer or vendor.
`document_sequences`	Next document number per type (e.g. INV-2026-0042).
`commissions`	Agent commission per deal, with the developer commission invoice.

8 Attendance and leave

6 tables

One attendance row per person per day with check-in photo. Corrections keep the original values. Approved leave cannot overlap and blocks new leads.

Example: Sara checks in at 9:02 with a photo. While she is on approved leave next week, she gets no new leads.

`attendance_records`	One row per employee per working day, with check-in/out photo, time and location.
`attendance_corrections`	Correction requests. The original values are kept in original_values; the record keeps the corrected values.
`leave_requests`	Leave requests and approvals. Approved leave makes the user unavailable for new leads.
`leave_types`	Kinds of leave and yearly allowance.
`attendance_policies`	Office hours, grace time, weekends and location rules.
`holidays`	Public and company holidays.

9 Subscriptions, billing and owner console

11 tables

How we sell the CRM. Each customer company has one subscription: plan, number of paid users (seats), monthly or yearly, card or bank transfer. We send VAT invoices; Stripe confirms card payments and the platform team confirms bank transfers. If payment fails, the company gets 14 days, then becomes read-only; data is never deleted while unpaid. The owner console is only for the platform team.

Example: Skyline Realty is on the Team plan with 10 seats at AED 99 per user. Adding an 11th active user is blocked until they buy a seat. If their card fails they have 14 days to pay, then the CRM becomes read-only. Their data is never deleted while unpaid.

`plans`	Subscription plans we sell (Solo, Team, Enterprise) and their prices.
`subscriptions`	Each tenant's current plan, seats, billing cycle and billing status (one per tenant).
`subscription_events`	History of every subscription change: trial, plan, seats, payment failure, read-only, reactivation, cancellation. Never changed.
`billing_invoices`	Our tax invoices to customers for their subscription (5% VAT), monthly or yearly.
`billing_invoice_lines`	Lines of a subscription invoice: seats x price, extra seats added mid-period (pro-rata), credits.
`billing_payments`	Card payments (from Stripe) and bank transfers (confirmed by the platform team) against subscription invoices.
`tenant_usage_daily`	Daily usage per tenant for the owner console: active users, leads, messages, storage.
`platform_admins`	People who run the CRM business (owner console). Not part of any tenant.
`platform_settings`	Business-wide settings: trial days, grace days, data retention, VAT rate.
`sales_enquiries`	Enterprise and "talk to sales" requests from the website.
`platform_audit_logs`	Every owner-console action (view tenant, suspend, extend trial, confirm transfer, change plans). Never changed.

10 Integrations, files, reports and audit

8 tables

Connected accounts (Meta, portals, WhatsApp, Google, Microsoft, webhook), every incoming webhook, sync runs, Meta reconciliation, files, exports and the audit log.

Example: The website form sends to the general webhook. The data is saved first, then turned into a lead, and retried automatically if anything fails.

`integrations`	Connected outside accounts (Meta, WhatsApp, portals, Google, Microsoft, general webhook). Tokens live in AWS SSM, not here.
`webhook_events`	Every incoming webhook, stored before processing so nothing is lost and duplicates are ignored.
`integration_sync_runs`	Each sync or import run with record counts and errors (shown on the Integrations screen).
`reconciliation_runs`	Meta lead reconciliation: leads at the source vs leads in the CRM, and missing IDs.
`files`	Every uploaded or generated file. The file itself is stored in S3.
`attachments`	Links a file to exactly one record: lead, project, unit, deal, financial document or commission.
`export_logs`	Every export: who, which report, filters, format, rows, when.
`audit_logs`	Who changed what and when, with before/after values. Never changed after saving.

For developers: column-by-column details, data types, keys, indexes and full ER diagrams are in CRM_Database_Design.pdf and in schema.sql, security.sql and seed.sql.